

Math 541

Week 2 Homework

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Exercise 1. Let $\alpha \in \mathbb{R}$. For $x \in \mathbb{R}^n$, define:

$$f(x) = \begin{cases} |x|^{-\alpha}, & 0 < |x| < 1 \\ 0, & \text{otherwise} \end{cases}$$

$$g(x) = \begin{cases} |x|^{-\alpha}, & |x| > 1 \\ 0, & \text{otherwise} \end{cases}.$$

For which α is $f \in L^1(\mathbb{R}^n)$? For which α is $f \in L^p(\mathbb{R}^n)$ for a given $1 \leq p < \infty$? Same questions for g .

Proof. We have:

$$\begin{aligned} \int_{\mathbb{R}^n} |f(x)|^p dx &= \int_{B(0,1)} |x|^{-\alpha p} dx \\ &= \int_0^1 \int_{S^{n-1}} r^{-\alpha p} r^{n-1} d\omega dr \\ &\leq C \int_0^1 r^{n-1-\alpha p} dr \\ &= C \lim_{\delta \rightarrow 0} \left(\frac{1}{n-\alpha p} - \frac{\delta^{n-\alpha p}}{n-\alpha p} \right). \end{aligned}$$

This limit converges only when $n - \alpha p > 0$. Thus $f \in L^p(\mathbb{R}^n)$ for $1 \leq p < \infty$ for all $\alpha < \frac{n}{p}$. Now consider:

$$\begin{aligned} \int_{\mathbb{R}^n} |g(x)|^p dx &= \int_{\mathbb{R}^n \setminus B(0,1)} |x|^{-\alpha p} dx \\ &= \int_1^\infty \int_{S^{n-1}} r^{-\alpha p} r^{n-1} d\omega dr \\ &\leq C \int_1^\infty r^{n-1-\alpha p} dr \\ &= C \lim_{\delta \rightarrow \infty} \left(\frac{\delta^{n-\alpha p}}{n-\alpha p} - \frac{1}{n-\alpha p} \right). \end{aligned}$$

This limit converges only when $n - \alpha p < 0$. Thus $g \in L^p(\mathbb{R}^n)$ for $1 \leq p < \infty$ for all $\alpha > \frac{n}{p}$. $\square$

Exercise 2. Let U be open in $\mathbb{R}^n$. Find a function $u \in L^1(U)$ which is unbounded on any open subset $V \subset U$. Hint: start with an enumeration of the points in U with rational coordinates. Then make us of translates of the function f (with an appropriate value for α) from the previous exercise.

Proof. Let f be given in the previous exercise for any $0 < \alpha < n$. Let $\mathbb{Q}^n \cap U = \{q_k \mid k \in \mathbb{N}\}$ be an enumeration of the rationals. Define:

$$u(x) := \begin{cases} \sum_{k=1}^{\infty} 2^{-k} f(x - q_k), & x \notin \mathbb{Q}^n \cap U \\ 0, & x \in \mathbb{Q}^n \cap U. \end{cases}$$

Since $u(x) = \sum_{k=1}^{\infty} 2^{-k} f(x - q_k)$ a.e., it is enough to show that the series of functions alone is in $L^1(U)$. Note that:

$$\int_U \sum_{k=1}^{\infty} 2^{-k} f(x - q_k) dx = \sum_{k=1}^{\infty} \int_U f(x - q_k) dx.$$

By the translating the function f , we have:

$$\begin{aligned} \int_U f(x - q_k) dx &= \int_{U - q_k} f(y) dy \\ &\leq \int_{\mathbb{R}^n} f(y) dy \\ &= \|f\|_{L^1(\mathbb{R}^n)}. \end{aligned}$$

Hence:

$$\begin{aligned} \int_U \sum_{k=1}^{\infty} 2^{-k} f(x - q_k) dx &\leq \sum_{k=1}^{\infty} 2^{-k} \|f\|_{L^1(\mathbb{R}^n)} \\ &= \|f\|_{L^1(\mathbb{R}^n)} \sum_{k=1}^{\infty} 2^{-k} \\ &< \infty. \end{aligned}$$

Thus $u \in L^1(U)$. Now let $V \subset U$ be any nonempty open set. Since $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$, there exists some $q_m \in V$. Since V is open, we can find points $x \in V \setminus \mathbb{Q}^n$ which are arbitrarily close to q_m . Hence:

$$\sum_{k=1}^{\infty} 2^{-k} f(x - q_k) \geq 2^{-m} f(x - q_m).$$

If $0 < |x - q_m| < 1$, then $f(x - q_m) = |x - q_m|^{-\alpha}$. This finally gives:

$$\sum_{k=1}^{\infty} 2^{-k} f(x - q_k) \geq 2^{-m} |x - q_m|^{-\alpha}.$$

As $x \rightarrow q_m$, the right hand side of the above inequality tends to infinity. Thus u is unbounded on V . $\square$

Exercise 3. Show that the fundamental solution Φ to Laplace's equation has a finite integral over any bounded neighborhood of the origin.

Proof. Let $U \subset \mathbb{R}^n$ be a bounded neighborhood of the origin. Then there exists $r > 0$ such that $U \subseteq B(0, r)$. Hence:

$$\int_U |\Phi(x)| \, dx \leq \int_{B(0, r)} |\Phi(x)| \, dx$$

For $n = 2$:

$$\begin{aligned} \int_{B(0, r)} |\Phi(x)| \, dx &= c_1 \int_{B(0, r)} |\log |x|| \, dx \\ &= c_1 \int_0^r \int_0^{2\pi} |r \log r| \, d\theta \, dr \\ &\leq C \int_0^r |r \log r| \, dr \\ &< \infty. \end{aligned}$$

For $n \geq 3$:

$$\begin{aligned} \int_{B(0, r)} |\Phi(x)| \, dx &= c_n \int_{B(0, r)} |x|^{2-n} \, dx \\ &= c_n \int_0^r \int_{S^{n-1}} r^{2-n} r^{n-1} \, d\omega \, dr \\ &\leq C \int_0^r r^{2-n} r^{n-1} \, dr \\ &= C \int_0^r r \, dr \\ &< \infty. \end{aligned}$$

Hence by our first inequality, the fundamental solution to Laplace's equation has a finite integral over U . $\square$

Exercise 4. For $i, j = 1, \dots, n$, calculate $\Phi_{x_i}(x)$ and $\Phi_{x_i x_j}(x)$ for any $n \geq 2$. Use your answer to show:

$$|D\Phi(x)| \leq \frac{C}{|x|^{n-1}} \text{ and } |D^2\Phi(x)| \leq \frac{C}{|x|^n}.$$

Proof. We will first show that $|D\Phi(x)| \leq \frac{C}{|x|^{n-1}}$. We proceed by cases on n . Case 1: $n = 2$. We have $\frac{\partial}{\partial x_i} \Phi(x) = -c_2 \frac{x_i}{|x|^2}$, hence $D\Phi(x) = -c_2 \frac{x}{|x|^2}$. Thus:

$$|D\Phi(x)| \leq c_2 \frac{|x|}{|x|^2} \leq \frac{C}{|x|}.$$

Case 2: $n \geq 3$. We have $\frac{\partial}{\partial x_i} \Phi(x) = c_n(2-n) \frac{x_i}{|x|^n}$, hence $D\Phi(x) = c_n(2-n) \frac{x}{|x|^n}$. Hence:

$$|D\Phi(x)| \leq c_n(2-n) \frac{|x|}{|x|^n} \leq \frac{C}{|x|^{n-1}}.$$

Using the above work, we can now show that $|D^2\Phi(x)| \leq \frac{C}{|x|^n}$. For $n = 2$:

$$\begin{aligned} \frac{\partial}{\partial x_j} \Phi_{x_i}(x) &= -c_2 \frac{\partial}{\partial x_j} \frac{x_i}{|x|^2} \\ &= -c_2 \frac{\partial}{\partial x_j} \frac{x_i}{|x|^2} \\ &= \delta_{ij} |x|^{-2} - 2x_i x_j |x|^{-4} \end{aligned}$$

For $n \geq 3$:

$$\begin{aligned} \frac{\partial}{\partial x_j} \Phi_{x_i}(x) &= -c_n(2-n) \frac{\partial}{\partial x_j} \frac{x_i}{|x|^n} \\ &= \delta_{ij} |x|^{-n} - nx_i x_j |x|^{-n-2}. \end{aligned}$$

In either case of dimension, we have:

$$\begin{aligned} |\delta_{ij} |x|^{-n}| &\leq \frac{1}{|x|^n}, \\ \left| \frac{x_i x_j}{|x|^{n+2}} \right| &\leq \frac{|x|^2}{|x|^{n+2}} = \frac{1}{|x|^n}. \end{aligned}$$

Thus:

$$|\Phi_{x_i x_j}(x)| \leq \frac{C}{|x|^n}.$$

Since every entry of the Hessian matrix is bounded by $\frac{C}{|x|^n}$, we have for $n \geq 2$:

$$|D^2\Phi(x)| \leq \frac{C}{|x|^n}. \quad \square$$

Exercise 5. Let $f \in C_c^2(\mathbb{R}^n)$ and $1 > \epsilon > 0$. Recall the following integrals that came up in our calculations in class:

$$\begin{aligned} I_\epsilon &= \int_{B(0,\epsilon)} \Delta_y f(x-y) \Phi(y) dy \\ J_\epsilon &= \int_{\partial B(0,\epsilon)} \frac{\partial f}{\partial \nu}(x-y) \Phi(y) dy. \end{aligned}$$

Show that for $n = 2$ we get:

- (a) $|I_\epsilon| \leq C\epsilon^2 |\log \epsilon|$.
- (b) $|J_\epsilon| \leq C\epsilon |\log \epsilon|$.

Proof. (a) We have:

$$\begin{aligned} |I_\epsilon| &= \left| \int_{B(0,\epsilon)} \Delta_y f(x-y) \Phi(y) dy \right| \\ &\leq \int_{B(0,\epsilon)} |\Delta_y f(x-y)| \Phi(y) dy \end{aligned}$$

Since $f \in C_c^2(\mathbb{R}^2)$, we have that $\Delta_y f(x-y)$ is bounded. Hence:

$$\begin{aligned} |I_\epsilon| &\leq C \int_{B(0,\epsilon)} \Phi(y) dy \\ &= C \int_{B(0,\epsilon)} -c_2 \log |y| dy \\ &\leq C \int_{B(0,\epsilon)} \log \left(\frac{1}{|y|} \right) dy \\ &= C \int_0^\epsilon \int_{S^1} r \log \left(\frac{1}{r} \right) d\omega dr \\ &\leq C \int_0^\epsilon r \log \left(\frac{1}{r} \right) dr. \end{aligned}$$

This is an improper integral at $r = 0$. After making the necessary adjustments, integration by parts yields:

$$\begin{aligned} C \lim_{\delta \rightarrow 0} \int_\delta^\epsilon r \log \left(\frac{1}{r} \right) dr &= C \lim_{\delta \rightarrow 0} \left(\frac{r^2}{2} \log \left(\frac{1}{r} \right) \Big|_\delta^\epsilon + \frac{1}{2} \int_\delta^\epsilon r dr \right) \\ &= C \lim_{\delta \rightarrow 0} \left(\frac{r^2}{2} \log \left(\frac{1}{r} \right) + \frac{r^2}{4} \right) \Big|_\delta^\epsilon \\ &= C \lim_{\delta \rightarrow 0} \left(\frac{\epsilon^2}{2} \log \left(\frac{1}{\epsilon} \right) + \frac{\epsilon^2}{4} - \frac{\delta^2}{2} \log \left(\frac{1}{\delta} \right) - \frac{\delta^2}{4} \right) \\ &= C \left(\frac{\epsilon^2}{2} \log \left(\frac{1}{\epsilon} \right) + \frac{\epsilon^2}{4} \right) \\ &= C \epsilon^2 \left(\frac{1}{2} |\log \epsilon| + \frac{1}{4} \right) \\ &\leq C \epsilon^2 \left(\frac{1}{2} |\log \epsilon| + \frac{1}{4} |\log \epsilon| \right) \\ &\leq C \epsilon^2 \left(\frac{1}{2} + \frac{1}{4} \right) |\log \epsilon| \\ &\leq C \epsilon^2 |\log \epsilon|. \end{aligned}$$

(b) Observe that:

$$\begin{aligned} |J_\epsilon| &= \left| \int_{\partial B(0,\epsilon)} \frac{\partial f}{\partial \vec{v}}(x-y) \Phi(y) dy \right| \\ &= \int_{\partial B(0,\epsilon)} \left| \frac{\partial f}{\partial \vec{v}}(x-y) \right| \Phi(y) dy \end{aligned}$$

Since $f \in C_c^2(\mathbb{R})$, we have that $\frac{\partial f}{\partial \vec{v}}(x-y)$ is bounded. Hence:

$$\begin{aligned} |J_\epsilon| &\leq C \int_{\partial B(0,\epsilon)} \Phi(y) dy = C \int_{\partial B(0,\epsilon)} -c_2 \log |y| dy \\ &\leq C \int_{\partial B(0,\epsilon)} \log \left(\frac{1}{|y|} \right) dy \\ &\leq C \int_{\partial B(0,\epsilon)} |\log |y|| dy \\ &\leq C |\log \epsilon| \int_{B(0,\epsilon)} dy \\ &= C |\log \epsilon| 2\pi\epsilon \\ &\leq C\epsilon |\log \epsilon|. \end{aligned}$$

□

Exercise 6. Recall the integral:

$$L_\epsilon = \int_{\partial B(0,\epsilon)} (D\Phi(y) \cdot \vec{v}) f(x-y) dS(y).$$

We showed that for $n \geq 3$ we get $L_\epsilon = \int f(x-y) dS(y)$. Show that we get the same thing for $n = 2$.

Proof. First consider:

$$\begin{aligned} \frac{\partial}{\partial y_i} \Phi(y) &= -c_2 \frac{\partial}{\partial y_i} \log |y| \\ &= -c_2 \frac{y_i}{|y|^2}. \end{aligned}$$

Hence:

$$D\Phi(y) = -c_2 \frac{y}{|y|^2}.$$

But since we're integrating on $\partial B(0, \epsilon)$, this means $D\Phi(y) = -c_2 \frac{y}{\epsilon^2}$. It remains to compute $D\Phi(y) \cdot \vec{v}$. Since we are on the ball of radius epsilon, the vector $\vec{v}$ is a function of the points

$y \in \partial B(0, \epsilon)$. In particular, $\vec{v} = \frac{y}{\epsilon}$. This gives:

$$\begin{aligned} D\Phi(y) \cdot \vec{v} &= -c_2 \frac{y}{\epsilon^2} \cdot \frac{y}{\epsilon} \\ &= -c_2 \frac{1}{\epsilon^3} (y \cdot y) \\ &= -c_2 \frac{1}{\epsilon^3} |y|^2 \\ &= -c_2 \epsilon^{-1}. \end{aligned}$$

Finally:

$$\begin{aligned} L_\epsilon &= \int_{\partial B(0, \epsilon)} -c_2 \epsilon^{-1} f(x - y) dS(y) \\ &= -c_2 \epsilon^{-1} \frac{2\pi\epsilon}{2\pi\epsilon} \int_{\partial B(0, \epsilon)} f(x - y) dS(y) \\ &= -c_2 \epsilon^{-1} 2\pi\epsilon \oint_{\partial B(0, \epsilon)} f(x - y) dS(y) \\ &= - \oint_{\partial B(0, \epsilon)} f(x - y) dS(y). \end{aligned} \quad \square$$

Exercise 7. Let $\epsilon > 0$.

(a) For $x \in \mathbb{R}$, let:

$$H(x) := \begin{cases} 0, & x < 0 \\ 1, & x \geq 0. \end{cases}$$

Compute the function $\zeta := \eta_\epsilon * H$. We call ζ the cutoff function.

- (b) Let $a < b$ be real numbers. Find a smooth function which is identically 1 on $[a, b]$, and is identically 0 for $x \leq a - \epsilon$ and $x \geq b + \epsilon$.
- (c) Now let $U, V \subset \mathbb{R}^n$ be open sets with $V \subset \subset U$. Show that there is a smooth function on $\mathbb{R}^n$ which is identically 1 on V , and identically 0 outside U .

Proof. (a) We have:

$$\begin{aligned}
 \zeta(x) &= (\eta_\epsilon * H)(x) \\
 &= \int_{\mathbb{R}} \eta_\epsilon(x-y) H(y) \, dy \\
 &= \int_0^\infty \eta_\epsilon(x-y) \, dy \\
 &= \int_{-\infty}^x \eta_\epsilon(t) \, dt \\
 &= \begin{cases} 0, & x \leq -\epsilon, \\ \int_{-\epsilon}^x \frac{c}{\epsilon} \exp\left(\frac{1}{\left(\frac{t}{\epsilon}\right)^2 - 1}\right) \, dt, & -\epsilon < x < \epsilon, \\ 1, & x \geq \epsilon. \end{cases}
 \end{aligned}$$

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